

Tests for data-generating mechanisms

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```
library(tidyverse)
library(MASS)
library(knitr)
library(survival)
library(timereg)

rprojroot <- rprojroot::find_rstudio_root_file()
source(file.path(rprojroot, "R/utils.R"))
source(file.path(rprojroot, "R/generate_data.R"))
```

1 DNA pathways

Example 1

We generate a simple example.

- Sample size: $n = 1000$
- Number of DNA pathways: $p = 3$
- Prevalence of DNA pathways: $\pi_1 = 0.1$, $\pi_2 = 0.4$, $\pi_3 = 0.6$
- Correlation matrix:

$$\Sigma_D = \begin{pmatrix} 1 & 0 & 0.2 \\ 0 & 1 & 0.5 \\ 0.2 & 0.5 & 1 \end{pmatrix}$$

```
set.seed(2026)

n <- 1000
prev <- c(0.1, 0.4, 0.6)

Sigma <- matrix(c(1, 0, 0.2,
                  0, 1, 0.5,
                  0.2, 0.5, 1),
                3, 3)

dna1 <- simulate_dna_pathways(n, prev, Sigma)
head(dna1)
```

```
  D1 D2 D3
1  0  1  1
2  0  0  0
3  0  1  1
4  0  0  0
5  0  0  0
6  0  0  0
```

```
# Check prevalences
colMeans(dna1)
```

```
  D1    D2    D3
0.093 0.405 0.603
```

```
# Check dependence (Fisher's exact tests)
fisher.test(table(dna1$D1, dna1$D2))
```

Fisher's Exact Test for Count Data

```
data:  table(dna1$D1, dna1$D2)
p-value = 1
alternative hypothesis: true odds ratio is not equal to 1
```

```
95 percent confidence interval:
 0.6397796 1.6015359
sample estimates:
odds ratio
 1.016585
```

```
fisher.test(table(dna1$D1, dna1$D3))
```

Fisher's Exact Test for Count Data

```
data: table(dna1$D1, dna1$D3)
p-value = 5.081e-05
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 1.628539 4.962215
sample estimates:
odds ratio
 2.780409
```

```
fisher.test(table(dna1$D2, dna1$D3))
```

Fisher's Exact Test for Count Data

```
data: table(dna1$D2, dna1$D3)
p-value < 2.2e-16
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 2.674120 4.802858
sample estimates:
odds ratio
 3.573428
```

Example 2

We generate only uncorrelated variables.

- Sample size:

$$n = 1000$$

- Number of DNA pathways:

$$p_D = 3$$

- Prevalence of DNA pathways: $\pi_1 = 0.1$, $\pi_2 = 0.4$, $\pi_3 = 0.6$
- Correlation matrix:

$$\Sigma_D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

```
set.seed(2026)
Sigma0 <- diag(3)
dna0 <- simulate_dna_pathways(n, prev, Sigma0)

# Check prevalences
colMeans(dna0)
```

```
      D1      D2      D3
0.111 0.409 0.591
```

```
# Check dependence (Fisher's exact tests)
fisher.test(table(dna0$D1, dna0$D2))
```

Fisher's Exact Test for Count Data

```
data:  table(dna0$D1, dna0$D2)
p-value = 0.3061
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.5128426 1.2159048
sample estimates:
odds ratio
 0.7941041
```

```
fisher.test(table(dna0$D1, dna0$D3))
```

Fisher's Exact Test for Count Data

```
data:  table(dna0$D1, dna0$D3)
```

```

p-value = 0.8379
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.6966913 1.6274484
sample estimates:
odds ratio
 1.060629

```

```
fisher.test(table(dna0$D2, dna0$D3))
```

Fisher's Exact Test for Count Data

```

data: table(dna0$D2, dna0$D3)
p-value = 0.3954
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.6846429 1.1616203
sample estimates:
odds ratio
 0.8916356

```

Example 3

We simulate correlated variables grouped into blocks.

- Sample size: $n = 1000$
- Number of DNA pathways: $p_D = 20$
- Prevalence of DNA pathways: $\pi_{p_D} \sim \mathcal{U}(0.1, 0.6)$
- Block size: $b_D = 5$
- Number of blocks: $K_D = 4$
- Within-block correlations: $\rho_{D;k1} = 0.6$, $\rho_{D;k2} = 0.5$, $\rho_{D;k3} = 0.4$, $\rho_{D;k4} = 0.1$
- Between-block correlation: $\rho_{B_D} = 0.05$

```

set.seed(2026)
p <- 20

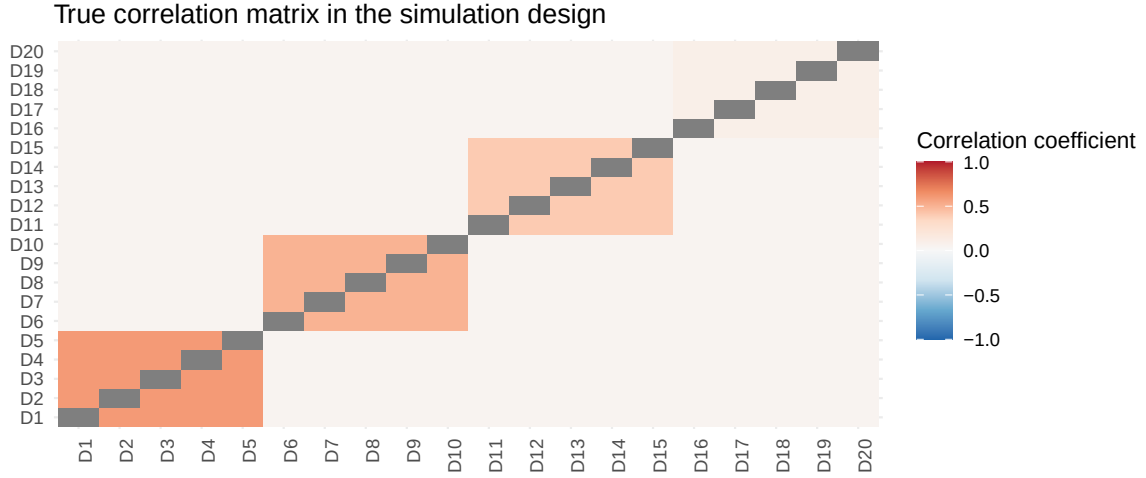
# Generate random prevalences
prev <- runif(20, 0.1, 0.6)
prev

```

```
[1] 0.4493367 0.3782653 0.1700700 0.2428617 0.3776845 0.1125656 0.3331153
[8] 0.5305053 0.2262506 0.3904032 0.1029666 0.4459330 0.2155625 0.5242662
[15] 0.1769176 0.2784047 0.3726023 0.1005988 0.2589122 0.1086447
```

```
# Initialize the matrix
Sigma <- generate_block_corr(p,
                             block_size = 5,
                             rho_within = c(0.6, 0.5, 0.4, 0.1),
                             rho_between = 0.05)

# Inspect the matrix
colnames(Sigma) <- paste0("D",1:p)
rownames(Sigma) <- paste0("D",1:p)
Sigma %>%
  as.data.frame()%>%
  tibble::rownames_to_column("var1")%>%
  pivot_longer(-var1, names_to = "var2")%>%
  mutate(value = ifelse(var1==var2, NA, value),
         var1 = factor(var1, levels = paste0("D",1:p)),
         var2 = factor(var2, levels = paste0("D",1:p)))%>%
  {
    ggplot(., aes(var1, var2, fill = value)) +
      geom_tile() +
      scale_fill_distiller(palette = "RdBu", direction = -1,
                          limits = c(-1,1)) +
      theme_minimal() +
      theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
      labs(fill = "Correlation coefficient", x = "", y = "",
           title = "True correlation matrix in the simulation design")
  }
```



```
# Simulate correlated data using this matrix
dna_block <- simulate_dna_pathways(n, prev, Sigma)

# Check marginal prevalences
data.frame(true_prev = prev,
            est_prev = colMeans(dna_block))%>%
  rownames_to_column("variable")%>%
  kable(format = "html", caption = "True and empirical prevalences",
        col.names = c("Variable", "True prevalence", "Estimated prevalence"))
```

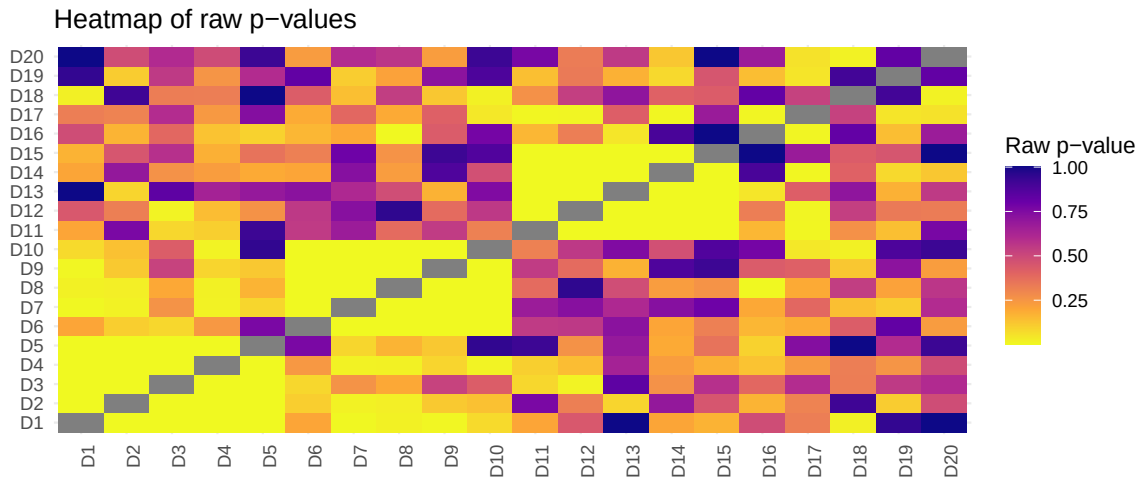
Table 1: True and empirical prevalences

Variable	True prevalence	Estimated prevalence
D1	0.4493367	0.463
D2	0.3782653	0.367
D3	0.1700700	0.161
D4	0.2428617	0.226
D5	0.3776845	0.371
D6	0.1125656	0.119
D7	0.3331153	0.298
D8	0.5305053	0.523
D9	0.2262506	0.199
D10	0.3904032	0.376
D11	0.1029666	0.119
D12	0.4459330	0.434
D13	0.2155625	0.207

Variable	True prevalence	Estimated prevalence
D14	0.5242662	0.522
D15	0.1769176	0.189
D16	0.2784047	0.287
D17	0.3726023	0.372
D18	0.1005988	0.105
D19	0.2589122	0.247
D20	0.1086447	0.122

```
# 20x20 matrix of raw p-values
pval_block <- pairwise_fisher(dna_block, names(dna_block))

# Heatmap of raw p-values
mat_long <- as.data.frame(as.table(pval_block$p.value.matrix))%>%
  mutate(Freq = ifelse(Var1==Var2, NA, Freq))
ggplot(mat_long, aes(Var1, Var2, fill = Freq)) +
  geom_tile() +
  scale_fill_viridis_c(option = "C", direction = -1) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(fill = "Raw p-value", x = "", y = "",
       title = "Heatmap of raw p-values")
```



2 RNA pathways

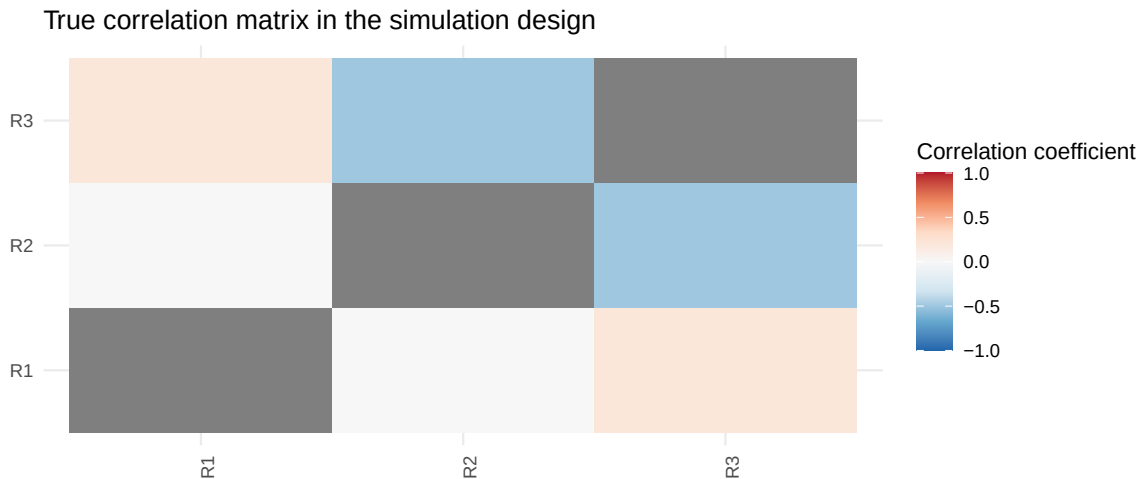
Example 1

We generate a simple example.

- Sample size: $n = 1000$
- Number of RNA pathways: $p_R = 3$
- Correlation matrix:

$$\Sigma_R = \begin{pmatrix} 1 & 0 & 0.2 \\ 0 & 1 & 0.5 \\ 0.2 & 0.5 & 1 \end{pmatrix}$$

```
set.seed(2026)
p <- 3
mu <- 0
Sigma <- matrix(c(1, 0, 0.2,
                  0, 1, -0.5,
                  0.2, -0.5, 1),
                3, 3)
# Inspect the matrix
colnames(Sigma) <- paste0("R",1:p)
rownames(Sigma) <- paste0("R",1:p)
Sigma %>%
  as.data.frame()%>%
  tibble::rownames_to_column("var1")%>%
  pivot_longer(-var1, names_to = "var2")%>%
  mutate(value = ifelse(var1==var2, NA, value),
         var1 = factor(var1, levels = paste0("R",1:p)),
         var2 = factor(var2, levels = paste0("R",1:p)))%>%
  {
    ggplot(., aes(var1, var2, fill = value)) +
      geom_tile() +
      scale_fill_distiller(palette = "RdBu", direction = -1,
                          limits = c(-1,1)) +
      theme_minimal() +
      theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
      labs(fill = "Correlation coefficient", x = "", y = "",
           title = "True correlation matrix in the simulation design")
  }
```



```
rna1 <- simulate_rna_pathways(n, p, mu, Sigma)
```

```
# Check mean
colMeans(rna1)
```

```
      R1      R2      R3
-2.063280e-17 -3.625572e-19 -1.562032e-17
```

```
# Check variance
apply(rna1, 2, var)
```

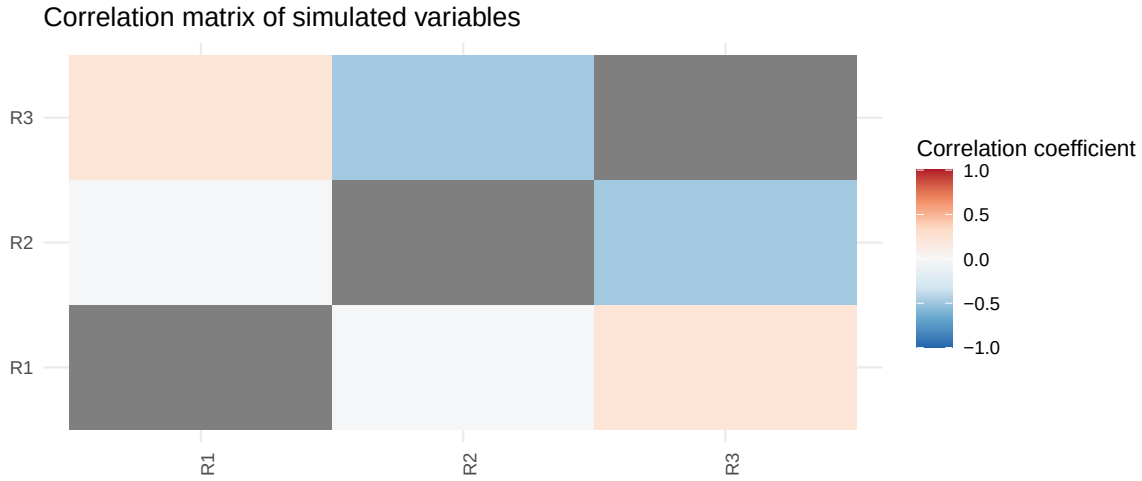
```
R1 R2 R3
1  1  1
```

```
# Correlation matrix
as.data.frame(cor(rna1)))>%
  tibble::rownames_to_column("var1")>%
  pivot_longer(-var1, names_to = "var2")>%
  mutate(value = ifelse(var1==var2, NA, value),
         var1 = factor(var1, levels = paste0("R",1:20)),
         var2 = factor(var2, levels = paste0("R",1:20)))>%
  {
    ggplot(., aes(var1, var2, fill = value)) +
      geom_tile() +
      scale_fill_distiller(palette = "RdBu", direction = -1,
```

```

limits = c(-1,1)) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
labs(fill = "Correlation coefficient", x = "", y = "",
      title = "Correlation matrix of simulated variables")
}

```



Example 2

We simulate correlated variables grouped into blocks.

- Sample size: $n = 1000$
- Number of RNA pathways $p_R = 20$
- Block size: $b_R = 5$
- Number of blocks: $K_R = 4$
- Within-block correlations: $\rho_{R;k1} = 0.7$, $\rho_{R;k2} = -0.2$, $\rho_{R;k3} = 0.5$, $\rho_{R;k4} = 0.1$
- Between-block correlation: $\rho_{B_R} = 0.05$

```

set.seed(2026)
p = 20

# Initialize the matrix
Sigma <- generate_block_corr(p,
                             block_size = 5,

```

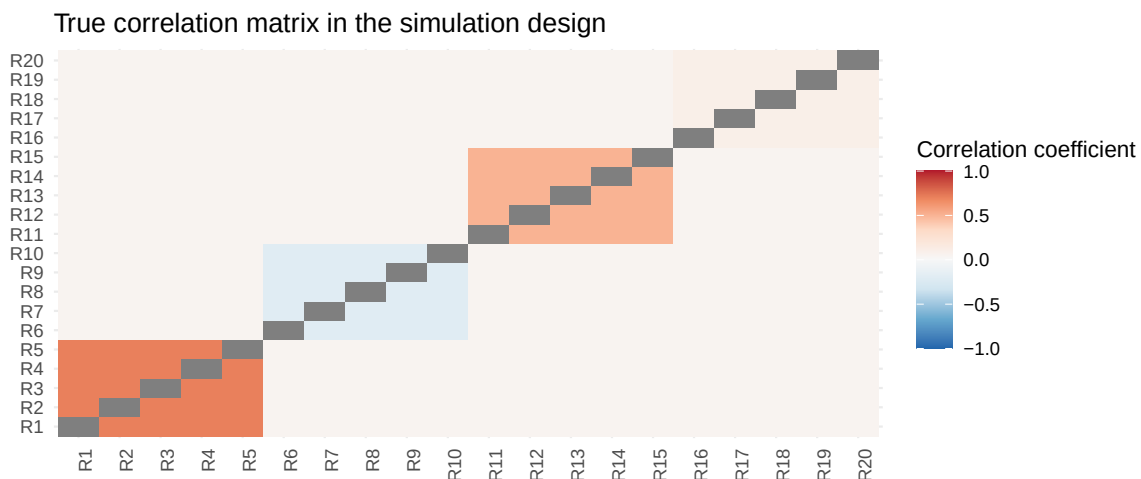
```

        rho_within = c(0.7, -0.2, 0.5, 0.1),
        rho_between = 0.05)

# Inspect the matrix
colnames(Sigma) <- paste0("R",1:20)
rownames(Sigma) <- paste0("R",1:20)
Sigma %>%
  as.data.frame()%>%
  tibble::rownames_to_column("var1")%>%
  pivot_longer(-var1, names_to = "var2")%>%
  mutate(value = ifelse(var1==var2, NA, value),
         var1 = factor(var1, levels = paste0("R",1:20)),
         var2 = factor(var2, levels = paste0("R",1:20)))%>%
  {
    ggplot(., aes(var1, var2, fill = value)) +
      geom_tile() +
      scale_fill_distiller(palette = "RdBu", direction = -1,
                          limits = c(-1,1)) +

      theme_minimal() +
      theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
      labs(fill = "Correlation coefficient", x = "", y = "",
           title = "True correlation matrix in the simulation design")
  }

```



```

# Simulate correlated data using this matrix
rna_block <- simulate_rna_pathways(n, p, mu, Sigma)

```

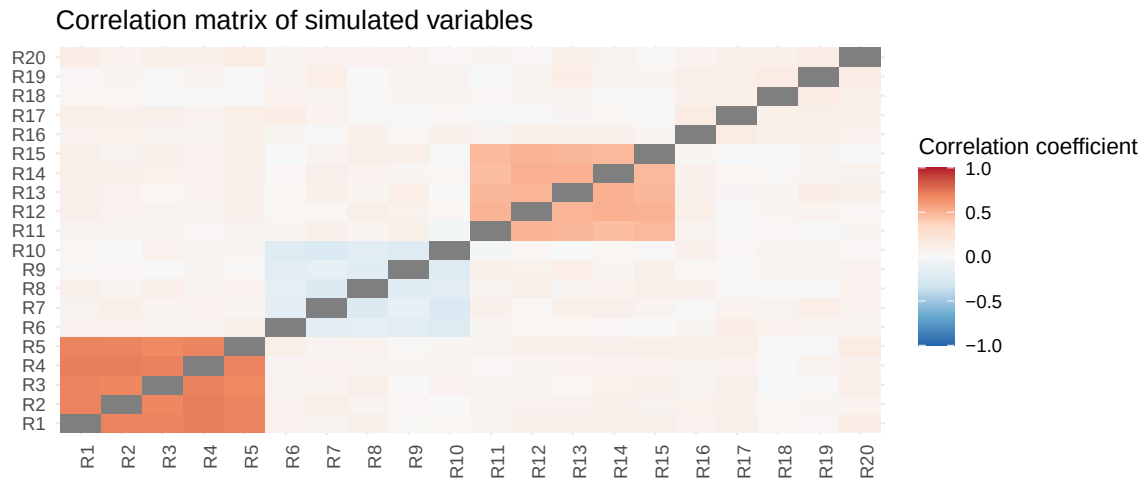
```
# Check mean
colMeans(rna_block)
```

	R1	R2	R3	R4	R5
	1.482950e-17	2.079370e-17	1.660477e-17	-2.885799e-17	2.405021e-17
	R6	R7	R8	R9	R10
	1.358288e-17	2.560452e-18	-1.540261e-17	-2.506675e-17	5.332540e-18
	R11	R12	R13	R14	R15
	3.231530e-17	1.650589e-18	1.955641e-17	-2.100750e-18	-1.665855e-17
	R16	R17	R18	R19	R20
	-1.226883e-17	-1.480554e-17	-1.961972e-17	-4.916206e-18	-4.945697e-18

```
# Check variance
apply(rna_block, 2, var)
```

R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

```
# Correlation matrix
as.data.frame(cor(rna_block))%>%
  tibble::rownames_to_column("var1")%>%
  pivot_longer(-var1, names_to = "var2")%>%
  mutate(value = ifelse(var1==var2, NA, value),
         var1 = factor(var1, levels = paste0("R",1:20)),
         var2 = factor(var2, levels = paste0("R",1:20)))%>%
  {
    ggplot(., aes(var1, var2, fill = value)) +
      geom_tile() +
      scale_fill_distiller(palette = "RdBu", direction = -1,
                          limits = c(-1,1)) +
      theme_minimal() +
      theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
      labs(fill = "Correlation coefficient", x = "", y = "",
           title = "Correlation matrix of simulated variables")
  }
```



3 Aalen model

```
# Set seed for reproducibility
set.seed(2026)

# Sample size
n <- 1000

# Generate covariates
p <- 3
Sigma <- matrix(c(1, 0, 0,
                  0, 1, 0,
                  0, 0, 1),
                3, 3)
X <- as.matrix(simulate_rna_pathways(n, p, mu = 0, Sigma))

# Generate time to event data
beta <- c(0, 0.02, 0.06)
lambda0 <- 0.07
lambda_i <- lambda0 + drop(X %*% beta)
lambda_i <- pmax(lambda_i, 1e-8)
T <- rexp(n, rate = lambda_i)
cens_rate <- 0.05
C <- rexp(n, rate = cens_rate)
time <- pmin(T, C)
```

```

event <- as.integer(T <= C)
data <- data.frame(time, event, X)

# Fit the Aalen model
# All components of the model have nonparametric time-varying effects.
aalen_fit <- timereg::aalen(Surv(time, event)~R1+R2+R3,
                           data = data)
summary(aalen_fit)

```

Additive Aalen Model

Test for nonparametric terms

Test for non-significant effects

	Supremum-test of significance	p-value	$H_0: B(t)=0$
(Intercept)	18.10	0.000	
R1	1.68	0.843	
R2	4.90	0.000	
R3	14.40	0.000	

Test for time invariant effects

	Kolmogorov-Smirnov test	p-value	H_0 :constant effect
(Intercept)	0.4020	0.240	
R1	0.0585	0.776	
R2	0.1750	0.309	
R3	0.3110	0.155	

	Cramer von Mises test	p-value	H_0 :constant effect
(Intercept)	2.0200	0.231	
R1	0.0476	0.636	
R2	0.3280	0.371	
R3	1.3100	0.135	

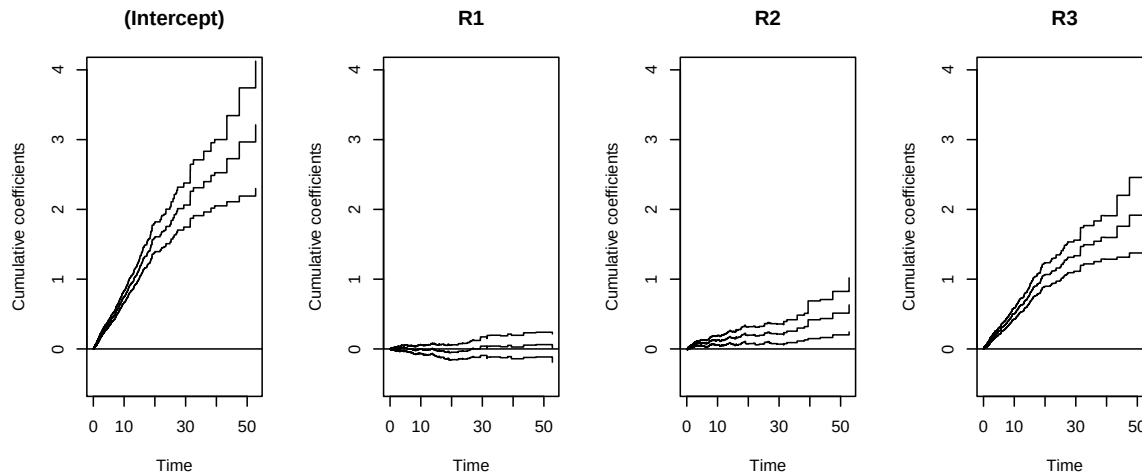
Call:

```
timereg::aalen(formula = Surv(time, event) ~ R1 + R2 + R3, data = data)
```

```

# Plot the non-parametric cumulative regression coefficients with 95% CI
par(mfrow=c(1,4))
plot(aalen_fit, ylim = c(-0.5,4))

```



```
# Fit the model with the effect of R1 being constant
aalen_fit2 <- timereg::aalen(Surv(time, event)~const(R1)+R2+R3,
                             data = data)
summary(aalen_fit2)
```

Additive Aalen Model

Test for nonparametric terms

Test for non-significant effects

	Supremum-test of significance	p-value	$H_0: B(t)=0$
(Intercept)	18.10	0	
R2	4.91	0	
R3	14.40	0	

Test for time invariant effects

	Kolmogorov-Smirnov test	p-value	$H_0: \text{constant effect}$
(Intercept)	0.387	0.293	
R2	0.190	0.280	
R3	0.301	0.191	

	Cramer von Mises test	p-value	$H_0: \text{constant effect}$
(Intercept)	1.820	0.271	
R2	0.417	0.325	
R3	1.090	0.209	

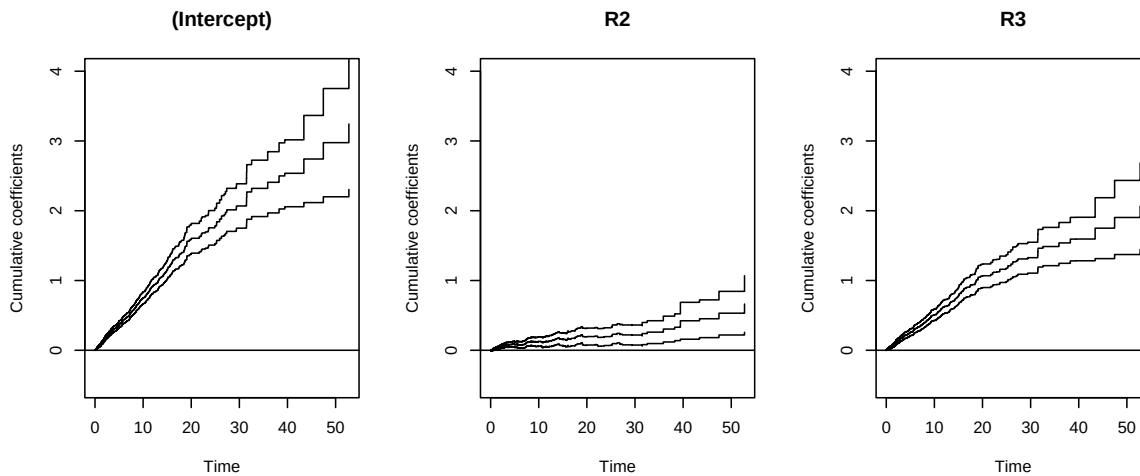
Parametric terms :

Coef.	SE Robust	SE	z	P-val	lower2.5%	upper97.5%
-------	-----------	----	---	-------	-----------	------------


```
const(R1) -0.000256 0.00217    0.00216 -0.119 0.906   -0.00451    0.004
```

```
Call:
timereg::aalen(formula = Surv(time, event) ~ const(R1) + R2 +
  R3, data = data)
```

```
par(mfrow=c(1,3))
plot(aalen_fit2, ylim = c(-0.5,4))
```



4 Cox-Aalen model

```
# Set seed for reproducibility
set.seed(2026)

# Sample size
n <- 1000

# Generate covariates
p <- 2
prev <- c(0.2, 0.5)
Sigma <- matrix(c(1, 0,
                  0, 1),
                p, p)
X_D <- simulate_dna_pathways(n, prev, Sigma)
```

```

X_R <- simulate_rna_pathways(n, p, mu = 0, Sigma)
X <- cbind(X_D, X_R)

# Generate time to event data
beta_add <- c(0, 0.04)
beta_mult <- c(0.7, 0)
lambda0 <- 0.05
cens_rate <- 0.05
hazard_i <- lambda0 * exp(drop(as.matrix(X_D) %*% beta_mult))+
  drop(as.matrix(X_R) %*% beta_add)
hazard_i <- pmax(hazard_i, 1e-6)
T <- rexp(n, rate = hazard_i)
C <- rexp(n, rate = cens_rate)
time <- pmin(T, C)
event <- as.integer(T<=C)
data <- data.frame(time, event, X_D, X_R)

# Fit Cox Aalen model
cox_aalen_fit <- timereg::cox.aalen(
  Surv(time, event)~prop(D1)+prop(D2)+R1+R2,
  data)
summary(cox_aalen_fit)

```

Cox-Aalen Model

Test for Aalen terms

Test not computed, sim=0

Proportional Cox terms :

	Coef.	SE Robust	SE	D2log(L) ⁻¹	z	P-val	lower2.5%
prop(D1)	0.7550	0.1090	0.110	0.1090	6.850	7.40e-12	0.541
prop(D2)	-0.0277	0.0927	0.093	0.0929	-0.298	7.66e-01	-0.209

upper97.5%

prop(D1) 0.969

prop(D2) 0.154

Test of Proportionality

	sup hat U(t)	p-value	H_0
prop(D1)	4.95	0.830	
prop(D2)	7.32	0.628	

```
# Proportional excess terms of model
cox(cox_aalen_fit)
```

Cox-Aalen Model

Test for Aalen terms
Test not computed, sim=0

Proportional Cox terms :

	Coef.	SE Robust	SE	D2log(L)^-1	z	P-val	lower2.5%
prop(D1)	0.7550	0.1090	0.110	0.1090	6.850	7.40e-12	0.541
prop(D2)	-0.0277	0.0927	0.093	0.0929	-0.298	7.66e-01	-0.209

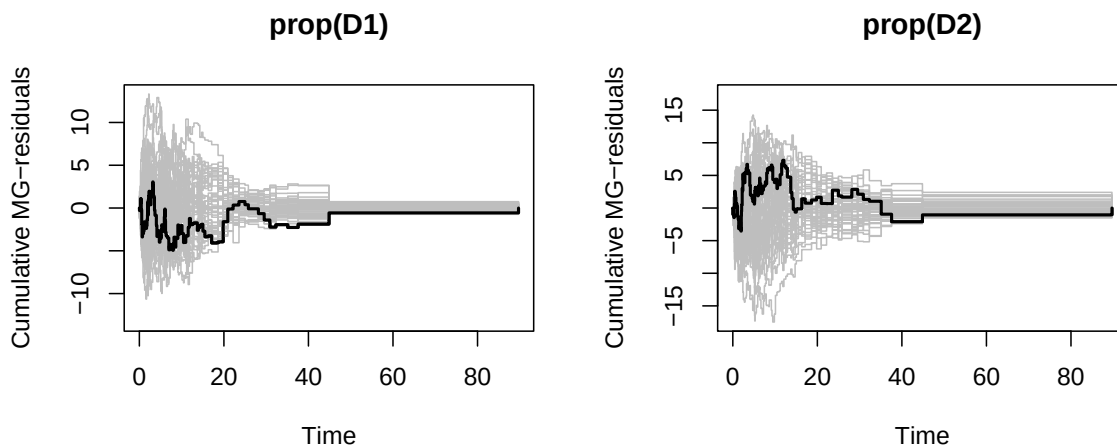
upper97.5%

prop(D1)	0.969
prop(D2)	0.154

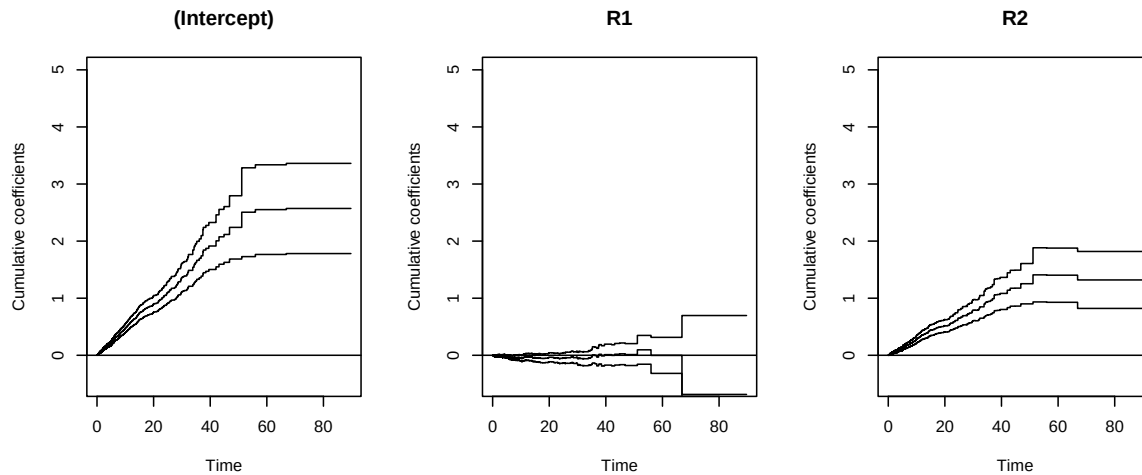
Test of Proportionality

	sup hat U(t)	p-value	H_0
prop(D1)	4.95	0.830	
prop(D2)	7.32	0.628	

```
par(mfrow=c(1,2))
plot(cox_aalen_fit,score= TRUE)
```



```
# Plot additive Aalen terms
par(mfrow = c(1,3))
plot(cox_aalen_fit, ylim = c(-0.5,5))
```



```
# Allow some additive terms to be time-constant
cox_aalen_fit2 <- timereg::cox.aalen(
  Surv(time, event)~prop(D1)+prop(D2)+const(R1)+R2,
  data, robust = TRUE)
summary(cox_aalen_fit2)
```

Cox-Aalen Model

Test for Aalen terms

Test not computed, sim=0

Proportional Cox terms :

	Coef.	SE Robust	SE	D2log(L) ⁻¹	z	P-val	lower2.5%
prop(D1)	0.7550	0.1090	0.110	0.1090	6.850	7.40e-12	0.541
prop(D2)	-0.0277	0.0927	0.093	0.0929	-0.298	7.66e-01	-0.209

upper97.5%

prop(D1) 0.969

prop(D2) 0.154

Test of Proportionality

	sup	hat U(t)	p-value	H ₀
prop(D1)		4.95	0.818	
prop(D2)		7.32	0.636	

```
# Proportional excess terms of model
```

```
cox(cox_aalen_fit2)
```

Cox-Aalen Model

Test for Aalen terms

Test not computed, sim=0

Proportional Cox terms :

	Coef.	SE	Robust SE	SE D2log(L)^-1	z	P-val	lower2.5%
prop(D1)	0.7550	0.1090	0.110	0.1090	6.850	7.40e-12	0.541
prop(D2)	-0.0277	0.0927	0.093	0.0929	-0.298	7.66e-01	-0.209

upper97.5%

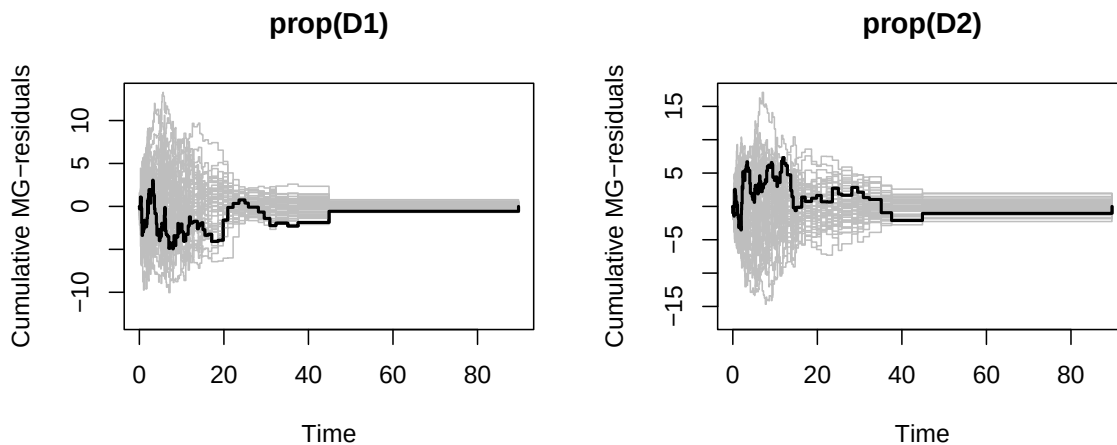
prop(D1) 0.969

prop(D2) 0.154

Test of Proportionality

	sup hat U(t)	p-value	H_0
prop(D1)	4.95	0.818	
prop(D2)	7.32	0.636	

```
par(mfrow=c(1,2))  
plot(cox_aalen_fit2,score= TRUE)
```



```
# Plot additive Aalen terms  
par(mfrow = c(1,3))  
plot(cox_aalen_fit2, ylim = c(-0.5, 5))
```

